

Block 3

Monetary Policy

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UNIT 7 CENTRAL BANK: ITS ROLE IN MONETARY POLICY

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7.0 OBJECTIVES

Having completed this unit, you will be able to:

- explain the concept of targets to be followed by the central banks;
- describe the various instruments of monetary policy;
- highlight the impact of global financial crisis on central banks;
- examine the use of unconventional instruments to tackle the challenges arising from financial crisis;
- compare the conventional and unconventional instruments in achieving the targets; and
- analyse the inflation targeting objective of RBI.

7.1 INTRODUCTION

Without doubt, money and monetary policy have evolved considerably. A monetary policy requires an arrangement of institutional practice to ensure that the right amount of money is made available to facilitate trade in an economy. It is no simple task to set the correct relative prices between various types of monies and goods and decide on how to get the right amount of money into circulation. Indeed, it can be argued that fiscal and monetary policy should *run in tandem*, so that in managing the latter attention is paid to the former and information is exchanged in the pursuit of the objectives jointly. It is worthwhile to recall that the nature and scope of the regulation of financial intermediation is closely linked to the monetary policy regime. In principle, there is a trade-off between designing instruments to stabilise the financial system and prevent excessively volatile financial outcomes, ensuring that financial sector retains appropriate incentives to create investment opportunities and allocate funds accordingly.

The ultimate decisions in policymaking may be based on judgements and that could, unfortunately remain faulty, even in the presence of right information and foresight. This unit provides information about various aspects of targets and instruments of monetary policy. It also discusses inflation targeting in India and reviews what RBI does to set targets and use of the instruments to achieve them.

7.2 TARGETS OF MONETARY POLICY

The formal mandate of the central bank differs significantly across countries. *De facto*, though, central banks across the world appear to be increasingly pursuing a single nominal target. By and large, this single nominal target is an inflation target. It is desirable for monetary policy to target something it can actually hope to deliver, on an average and over time. The question relating to the choice of appropriate target for conducting monetary policy goes into the basic question of the interrelationship between money, output and prices. In general, most central banks have a broad mandate that includes not only price (or exchange rate) stability, but also the safeguarding of financial stability, as well as the promotion of economic growth, and, sometimes, financial development. The growth objective has also come to the forefront. To fulfil these multiple objectives within the constraints imposed by a particular policy regime, central banks use all available tools, including cooperation in some instances with fiscal authorities and other policymakers. Importantly, interest rates and foreign exchange intervention may not be sufficient to resolve the policy dilemmas resulting from the impact of external factors on the domestic financial system.

There is a two-way street of monetary policy operating through financial markets and financial markets feeding back into monetary policy. The problem of central bank is compounded by the fact that their instruments do not directly affect the goals. The instruments affect variables such as money

supply and interest rates, which then affect goal variables with a lag. In addition, these lags may be uncertain. Due to these problems, in the conduct of monetary policy, a distinction is made among:

- i) goals (or objectives),
- ii) targets (or intermediate targets),
- iii) indicators (or operational targets), and
- iv) instruments (or tools).

Target and indicator variables lie between goal and instrument variables. Target variables such as money supply and interest rates have direct and predictable impact on goal variables and can be quickly and more easily observed. Target variables are not directly affected by central bank instruments. The instruments affect target variables, through another set of variables called indicators. These indicators such as monetary base and short run interest rates are more responsive to instruments.

The conduct of monetary policy can be represented schematically as follows:

Instruments → Indicators → Targets → Goals

Following is the list of different kinds of variables.

Table 7.1: Kinds of variables

Goals or Objectives

- i) High Employment.
- ii) Economic Growth.
- iii) Price Stability.
- iv) Interest-Rate Stability.
- v) Stability of Financial Markets.
- vi) Stability in Foreign Exchange Markets.

Targets or Intermediate Targets

- i) Monetary Aggregates (M1, M2, M3 etc.).
- ii) Short Run and Long Run Interest Rates.

Indicators or Operational Targets

- i) Monetary Base or High-Powered Money.
- ii) Short Run Interest Rate (Rate on Treasury Bill, Overnight Rate).

Instruments or Tools

- i) Open Market Operations.

- ii) Reserve Requirements.
- iii) Operating Band for Overnight Rate.
- iv) Bank Rate.

(Source: <http://web.uvic.ca/~kumara/econ506/lecture4.pdf>)

Despite listing six goals, it does not mean that different countries and regimes give same weight to all these goals. Different goals may get different emphasis in different countries and over times. All the goals may not be compatible with each other. For instance, goal of price stability may conflict with the goals of high employment and stability of interest rate at least in the short run.

Three criteria determine how the target variables are chosen. These are: (i) measurability, (ii) controllability, and (iii) predictable effects on goals. Quick and accurate measurement of target variables is necessary because the target will be useful only if it signals rapidly when policy is off-track. For a target variable to be useful, a central bank must be able to exercise effective control over it. If the central bank cannot exercise effective control over it, knowing that it is off-track is of little help. Finally, and most importantly, target variables must have a predictable impact on goal variables. If target variables do not have predictable impact on goal variables, the central bank cannot achieve its goal by using target variables. Monetary aggregates and short and long run interest rates satisfy all three criteria. The same three criteria are used to choose indicators. They must be measurable, the central bank should have effective control over them, and they must have predictable effect on target variables. The general conclusion is that if the main source of disturbance in the economy is shocks to goods market (IS curve), then targeting money supply (or using money supply tool) is optimal. On the other hand, if the main source of disturbance is shocks to demand for money or financial market (LM curve), then targeting interest rate is optimal.

Check Your Progress 1

- 1) What do you mean by targets of Monetary Policy ?

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- 2) Distinguish between following elements of Monetary Policy: goals (objectives), intermediate targets, indicators (or operational targets), and instruments.

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3) What are the objectives of monetary policy?

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7.3 INSTRUMENTS OF MONETARY POLICY

The lesson learnt from the global financial crisis has led central bankers to delve into uncharted territory adopting hitherto unheard-of unconventional policies. Central banks have rediscovered their role as the *Lender of Last Resort* (LoLR). The current trend worldwide and the present debate within the country suggests that the end of stratification between sectors and consolidation within sub-sectors would be inevitable. The complexity of the transactions and large funding requirements demand an innovative approach towards financial structuring and the use of a variety of financial instruments. However, suitable prudential norms must be in place and adhered to, for the process to function smoothly. Only central banks have the requisite market pulse and instruments to make contextual judgments.

7.3.1 Conventional Instruments

When a nation's economy becomes "overheated" – growing rapidly to the point that inflation increases to uncomfortable levels – the central bank enacts restrictive monetary policy to tighten the money supply. This effectively reduces the amount of money in circulation and also the rate at which new money enters the system. Raising the target interest rate makes money more expensive and increases borrowing costs, reducing the demand for cash and cash instruments. The bank may increase the level of reserves that commercial and retail banks must keep on hand, limiting their ability to generate new loans. The central bank can also sell government bonds from its balance sheet in the open market, exchanging those bonds by taking in money from circulation. When a nation's economy slips into recession, these policy tools can be operated in reverse, constituting a loose or expansionary monetary policy. Interest rates are lowered, reserve limits loosened, and instead of selling bonds in the open market, they are purchased in exchange for newly created money.

7.3.2 Unconventional Instruments

According to *Deepak Mohanty* of RBI, "when central banks look beyond their traditional instrument of policy interest rate, monetary policy takes an unconventional character." An unconventional policy instrument is defined as any instrument of the central bank that does not depend for its operation on changing the risk-free nominal interest rate now or in the future. Unconventional monetary policy spans from quantitative easing to debt monetisation. In other words, the term "unconventional" means policy instruments other than standard tools such as interest rates or foreign exchange market intervention. These unconventional measures are:

- i) liquidity provision to banks on extraordinary terms, especially at longer maturities, in order to alleviate pressures in the interbank market;
- ii) intervention in selected credit markets to support secondary market liquidity and credit pricing and supply; and
- iii) outright bond purchases aimed at easing financing conditions beyond what could be achieved by policy rate cuts.

The rationale for the use of unconventional Central bank policy instruments is fairly straight forward. One can group these unconventional instruments into three broad categories:

- i) unconventional monetary policy and macro-prudential tools aimed at offsetting the financial implications of capital flows;
- ii) balance sheet policies other than foreign exchange intervention; and
- iii) fiscal and quasi-fiscal measures to offset the domestic consequences of foreign exchange intervention.

Following are the main policy tools:

- i) quantitative easing (QE);
- ii) forward guidance;
- iii) negative interest rates;
- iv) overt monetary financing of fiscal deficits; and
- v) debt monetisation.

QE has been crucial to avert financial meltdown, and has had some effects on the macro-economy. Yet there are reasons to be critical about its effectiveness for macroeconomic stabilisation relative to its costs and risks. QE has dramatically increased base money, but not the money supply that drives aggregate demand. A major factor conditioning the macroeconomic effects of QE is its interaction with fiscal policy. Pulling QE alongside a restrictive fiscal policy is like pushing on the car's accelerator and brake pedals at the same time, which is what has happened in the US and UK.

So-called '*helicopter money*' policies – those in which government spending or transfers to households are paid for by printing money – involve both monetary and fiscal policy. This means they require extraordinary cooperation between the government and the central bank, which potentially undermines central-bank independence. However, emergency policies of this type may be justified during extreme systemic crises. Injections of helicopter money can increase net wealth and thus stimulate spending, and this

mechanism is particularly important when conventional monetary policy is stuck at the zero lower bound.

Table 7.2 offers a snapshot of what is considered to be the main features of each policy type. In the table 7.2, policies are reported from left to right in ascending order of the directness of their impact on spending – from those that rely on changes in prices and expectations to those that affect spending by adding money balances to the economy.

Table 7.2: Unconventional monetary policies: A synopsis

Policy Types Indicators	FG	NIR	QE	DM	OMF	NC
Impact on aggregate demand: - Transmission channel - Effectiveness	Indirect (via expectations on future interest rates) Short gestation Limited by ZLB Uncertain impact	Indirect (via interest rates) Short gestation Poor public acceptance Strong and quick impact	Indirect (via assets prices and inflation expectations) Short gestation Slow and moderate impact	Indirect (via regained fiscal space) Long gestation Quick, strong, and long lasting impact	Direct (via helicopter money balances) Long gestation Quick and strong impact	Direct (via helicopter money balances) Long gestation Quick and strong impact
Central bank independence	Unhindered	Unhindered	Unhindered	Central bank /Government cooperation required	Central bank /Government cooperation required	Central bank not involved in complementary money policy

FG = forwarding guidance

NIR = negative interest rate

QE = quantitative easing

DM = debt monetisation

OMF = overt monetary financing of fiscal deficits with extreme neo-chartlist form

NC = neo-chartalism

(Source: <https://voxeu.org/article/unconventional-monetary-policies-revisited-part-ii>)

The policies that have greater direct impact on spending (overt monetary financing, neo-chartalism) are those that combine expansionary fiscal impulses with permanent monetary financing (“helicopter money”). This combination requires a degree of cooperation between the government and the central bank, with implications for central-bank independence. Government (and political) involvement, as well as the necessary coordination with the central bank, entail longer policy gestation periods than for policies involving the central bank exclusively (forward guidance, negative interest rates, and QE). On the other hand, the transmission from the fiscal plus-monetary policy impulse to the spending response – which is inherent in helicopter money options – is more direct, quicker and stronger.

Effective macro-prudential regulation and supervision is being devised to take care of systemic risks that lead to such financial instability. The macro-prudential approach focuses not only on individual balance sheets but also interconnectedness and the network externalities that they generate. This is expected to take care of systemic risks. The macro prudential approach to regulating institutions and markets is now a key plank of central banks’ efforts towards maintaining financial stability.

This includes several types of balance-sheet policies where the central bank purchases assets (for example through quantitative easing or credit easing), emergency lending, and subsidized access to central-bank credit. Unconventional policy as defined also includes ‘macro-prudential’ policy instruments that place restrictions on borrowing independently of interest rates.

QE injects money to owners of assets (e.g., traders, hedge funds, investors, banks, high wealth individuals, and speculators) who benefit from QE-stirred bond and asset-price rises, but represent a tiny minority with a low propensity to consume. Conversely, QE does not reach common people (with higher propensity to consume). Indications are that QE has significant distributional effects.

7.3.3 Comparing Conventional and Unconventional Instruments

It is important to have a good understanding of the unconventional policies and how they differ from the conventional ones. Overall non-conventional tools would seem to have some financial stability considerations. According to *Lorenzo Bini Smaghi* (Member of the Executive Board of the European Central Bank) the decision whether to embark on non-conventional monetary policy measures, which specific ones and for how long depends on a series of considerations, including the overall conditions of the economy, in particular the presence of deflationary pressures and the effective constraint provided by the zero lower bound for interest rates, the structure of the financial flows in the economy, the medium term incentives that would be created for the private sector and the risks entailed in the exit strategy, including for the independence and credibility of the central bank. Some measures present

greater advantages, in terms of providing the right incentives and being easily reversible. Others present more risks and have to be well targeted and used under specific circumstances.

7.3.4 The Transmission Mechanism

The relationship between the central bank's instrument and the target variables is often referred to as the monetary policy transmission mechanism. In fact, this also involves a number of other economic variables and relations. The short-term interest rate determined by the central bank and the target variables (e.g. the exchange rate or inflation and the output gap) will be included along with other macroeconomic variables such as consumption, investment, government budget balance, balance of payments, long-term interest rate and relevant foreign variables. Influences from abroad may also lead to shifts in the transmission mechanism. The relationship between the monetary policy instrument and the targets is not invariant, and the central bank must therefore be ready to adjust its monetary policy if the relationship changes. Since fiscal policy plays an important role in the central bank's ability to reach its targets, adjustment of fiscal policy may be necessary if monetary policy alone cannot ensure that the monetary policy targets are met.

Dynamic transmission: If changes to the monetary policy instrument have a delayed effect on the monetary policy targets, the central bank's task becomes considerably more difficult. When monetary policy is based on flexible inflation targeting, the relationship is dynamic since the target is to control inflation in the medium term. The central bank changes its short-term interest rate in order to influence inflation e.g. two years ahead. This means that the central bank cannot immediately assess whether an adjustment of the monetary policy instrument has the desired impact on its target variable. It is clear that the horizon for the monetary policy target is of major importance to the "to technique" of the central bank's implementation of monetary policy and to its external communication.

Stochastic transmission: This transmission mechanism has been described as a deterministic relation. In practice, however, the transmission from the monetary policy instrument to the targets is subject to uncertainty. When the central bank changes its short-term interest rate, the effect on the exchange rate or the future inflation is not known precisely. The importance of this depends on whether the relationship between the instrument and the targets is dynamic. The longer and more complex the transmission mechanism, the greater the risk that e.g. unexpected shocks to the economy prevent the central bank from meeting its targets.

7.4 GLOBAL FINANCIAL CRISIS AND CENTRAL BANKS

In response to the global financial crisis central banks have embarked on a variety of unconventional monetary policies. Attempts were made to make money easier and cheaper through an increase in supply of money and credit